

Digital Forensics Fundamentals

Module II

Introduction of Digital Forensics

- Digital forensics technically involves gathering evidence from any digital device.
- Digital Forensics is the process of identifying, preserving, analyzing and presenting digital evidences
- To ensure that criminal investigations handled digital evidence in a way that was admissible in court, officials established specific procedures.

Need for Digital Forensics

Digital forensics is needed because of the rapid growth of computers, mobile devices, and the internet, which has led to an increase in cybercrimes.

Increase in Cybercrimes: Crimes like hacking, identity theft, online fraud, cyberstalking, and data breaches require technical investigation.

Evidence in Digital Form: Modern crimes leave digital traces such as emails, logs, files, browser history, and messages.

Legal and Court Requirements: Courts require reliable and tamper-proof digital evidence that follows proper procedures.

Corporate and Organizational Security: Organizations need forensic analysis to detect insider threats, data leaks, and policy violations.

Incident Response and Recovery: Helps understand how an attack occurred and prevents future incidents.

Objectives of Digital Forensics

- **Evidence to Court:** It recovers, analyzes, and preserves digital and forensic evidence to help the department's investigation to present the evidence in court.
- **Identifying the Culprit:** It aims to cause the attacks and identify the main culprit behind the crimes.
- **Legal Procedures:** To ensure the evidence found at a suspicious crime scene is uncorrupted, we design the methods for collecting and preserving the evidence.
- **Data Redundancy:** Recover the deleted files and subdivide them from digital media to validate them.

Types of Digital Forensics

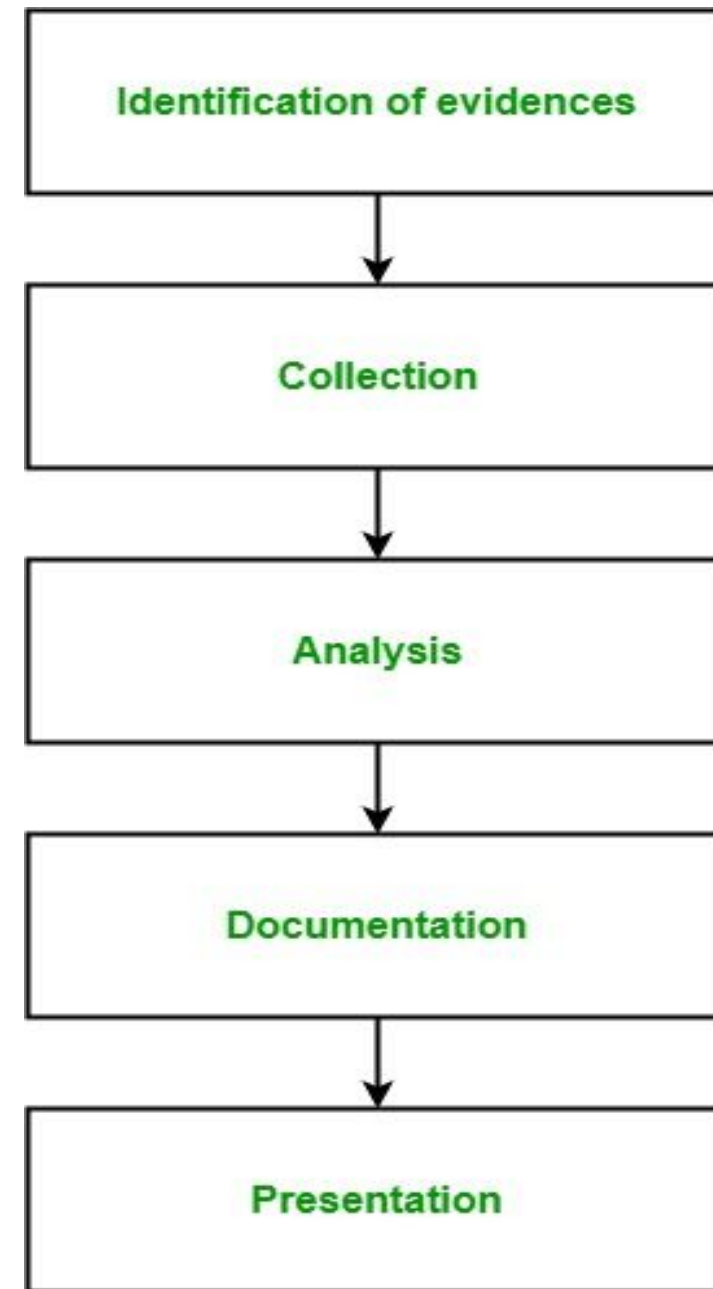
Computer Forensics: identifying, acquiring, and preserving evidence to show facts is part of it.

Network Forensics: looks at data that is being sent from one network to another. Network data is transient and must be captured in real time. Investigators record, study, and write down what's happening on a network in order to find online threats or unauthorized access.

Mobile Device Forensics: get data back from broken or lost smartphones, tablets, GPS devices, and other portable electronics.

Database Forensics: To find signs of fraud, tampering, or changes that were not allowed.

Process Digital Forensic



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- 1. Identification of evidence:** Related to the digital crime in storage media, hardware, operating system, network and/or applications.
- 2. Collection:** Doesn't degrade to vanish with time. Preserving the digital evidences is very important and crucial.
- 3. Analysis:** Analyzing the collected digital evidences of the committed computer crime in order to trace the criminal and possible path used to breach into the system.

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4. Documentation: It includes the proper documentation of the whole digital investigation, digital evidences, loopholes of the attacked system etc. so that the case can be studied and analysed in future also and can be presented in the court in a proper format.

5. Presentation: It includes the presentation of all the digital evidences and documentation in the court in order to prove the digital crime committed and identify the criminal.

Benefits of Digital Forensics

Cybercrime Investigation: helps identify the attacks, their methods, and the extent of the damage caused. By doing so, law enforcement agencies can take appropriate actions to bring criminals to justice and prevent future attacks.

Evidence Collection and Preservation: strengthens the legal proceedings against cybercriminals and helps in securing convictions.

Data Recovery: identify and address intellectual property theft, safeguarding a company's competitive edge and proprietary information.

Incident Response and Mitigation: cyberattack or security breach, digital forensics enables organizations to respond promptly and effectively. By analyzing the attack vector and identifying vulnerabilities

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Intellectual Property Protection: help identify and address intellectual property theft, safeguarding a company's competitive edge and proprietary information.

Employee Misconduct Investigations: Digital forensics can be employed to investigate employee misconduct, such as data theft, unauthorized access, or policy violations.

Enhancing Cybersecurity Practices: By analyzing past cyber incidents, organizations can gain insights into potential weaknesses in their cybersecurity infrastructure.

Digital Policy Compliance: Various industries and organizations have specific regulations and policies related to digital data handling and privacy

Public Safety and National Security: In cases related to terrorism, organized crime, or other national security threats, digital forensics is invaluable in gathering intelligence and preventing potential harm to the public.

Chain of Custody

WHAT IS CHAIN OF CUSTODY & EVIDENCE HANDLING?

As electronic evidence is easy to tamper or to get damaged, it is necessary for the court to know exactly who, what, when, where, and why was the evidence transferred to the concerned person.

It will not be possible to prove the integrity of the evidence, if the chain of custody is not properly maintained.



- Chain of custody refers to the documentation that shows the people who have been entrusted with the evidence.
- These would be -
 1. People who have seized the equipment
 2. People who are in charge of transferring the evidence from the crime scene to the forensic labs.
 3. People in charge of analysing the evidence, and so on.

Chain of Custody

Indicates the collection, sequence of control, transfer, and analysis.

Also documents each person who handled the evidence, the date/time it was collected or transferred, and the purpose for the transfer.

Important to maintain the chain of custody to preserve the integrity of the evidence and prevent it from contamination, which can alter the state of the evidence.

If not preserved, the evidence presented in court might be challenged and ruled inadmissible.

Procedure to establish the chain of custody

You should ensure the following procedure is followed according to the chain of custody for electronic evidence:

- **Save the original materials:** You should always work on copies of the digital evidence as opposed to the original. This ensures that you are able to compare your work products to the original that you preserved unmodified.
- **Take photos of physical evidence:** Photos of physical (electronic) evidence establish the chain of custody and make it more authentic.
- **Take screenshots of digital evidence content:** In cases where the evidence is intangible, taking screenshots is an effective way of establishing the chain of custody.

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- **Document date, time, and any other information of receipt:** Recording the timestamps of whoever has had the evidence allows investigators to build a reliable timeline of where the evidence was prior to being obtained. In the event that there is a hole in the timeline, further investigation may be necessary.
- **Inject a bit-for-bit clone of digital evidence content into our forensic computers:** To ensure that we obtain a complete duplicate of the digital evidence in question.
- **Perform a hash test analysis to further authenticate the working clone:** Performing a hash test ensures that the data we obtain from the previous bit-by-bit copy procedure is not corrupt and reflects the true nature of the original evidence.

Digital Evidence

Digital evidence is any information of probative value that is **stored, processed, or transmitted in digital form** and can be used in an investigation or court of law.

Commonly used in **cybercrime investigations, criminal cases, corporate fraud, and digital forensics.**

Analysis of electronic evidence extracts relevant information, such as tracing communication patterns or identifying unauthorized access.

Legal compliance, including adherence to the chain of custody protocols, is crucial to maintaining its admissibility in court.

Ultimately, electronic evidence is used in understanding cyber incidents and extracting hacking effectively.

Types of Digital Evidence

Document Files

Emails

Social Media Content

Multimedia Files

Web Browsing History

Database Records

Sources of Digital Evidence

Computers and Laptops: Personal computers and laptops are primary sources of digital evidence. They store documents, emails, browsing history, and even deleted files.

Mobile Devices: With the prevalence of smartphones and tablets, these devices have become prominent sources of digital evidence. Text messages, call logs, GPS data, and app usage can all be extracted and analyzed.

Cloud Services: Cloud storage and services like Google Drive, Dropbox, and iCloud can also hold valuable evidence. Information stored in the cloud might include documents, images, and backups of mobile devices.

Surveillance Cameras: In both public and private settings, surveillance cameras capture video evidence. Footage from these cameras can be pivotal in criminal investigations.

IoT Devices: Introduced a multitude of devices that can serve as sources of digital evidence. Smart home devices, wearables, and even connected appliances can offer insights into individuals' actions and behaviors.

Network Logs: Maintained by ISPs or organizations can provide a trail of digital evidence. They may contain records of internet activity, connections, and traffic.

Online Services: Information stored on online platforms, such as social media, email services, and cloud storage, can be requested as digital evidence when relevant to an investigation.

Server Logs: For businesses and websites, server logs can reveal patterns of usage, access, and any unauthorized activity.

Rules of Digital Evidences

Obtain explicit permission: Before performing any penetration tests or security assessments, obtain explicit written consent. This guarantees legality and establishes the extent.

Define the scope clearly: Recognize and accept the precise networks, applications, or systems that will be put to the test. Don't cross these lines.

Report all findings transparently: All vulnerabilities found should be clearly and thoroughly documented and communicated to the customer or organization.

Not exploit vulnerabilities for personal gain: The goal is to find vulnerabilities so they can be fixed, not to use them to gain illegal access or make money.

Minimize Harm: During the testing process, take care to prevent any harm, interruption, or data loss.

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Respect confidentiality: Maintain the highest level of confidentiality for all information acquired during the engagement and refrain from disclosing it without authorization.

Erase traces of testing: After the evaluation is over, restore the systems to their initial configuration by deleting any tools, scripts, or backdoors that were utilized.

Act professionally and ethically: Throughout the interaction, remain extremely professional and uphold moral standards.

Continuously improve skills and knowledge: Keep abreast of the most recent dangers, weaknesses, and ethical hacking methods.

Operate within the law: Respect all applicable cybersecurity and data privacy laws and rules in the operating jurisdiction (currently New Delhi, Delhi, India).

Incident Response

Organization must have Strategic, organized respond an organization uses a cyberattack.

The response is executed according to planned procedures that seek to limit damage and repair breached vulnerabilities in systems.

Professionals use incident response plans to manage security incidents.

Having a clearly defined incident response plan can limit attack damage, lower costs, and save time after a security breach.

CSIRT: Computer Security Incident Response Team

Formed by the people responsible for leading or handling the response to an incident.

The team is crucial to running incident response exercises, providing staff training, and maintaining security awareness.

CSIRT involves several core roles, which can be played by one or more people.

These include **senior** and **executive management**, who are responsible for making critical decisions, and an **incident manager**, who ensures all actions are tracked and the incident is clearly documented, communicated to **stakeholders**, and escalated.

The CSIRT also includes **leaders** from customer service, **human resources**, **legal**, and **public relations departments**.

It requires **analysts**, **investigators**, and **IT infrastructure experts**, who will typically be from an external organization, to explore, contain, and remediate the incident

Incident Response Plan Process



Preparation

Determines how well an organization will be able to respond in the event of an attack.

It requires several **key elements** to have been implemented to enable the organization to handle an incident:

Policy

Response plan/strategy

Communication

Documentation

Team: CSIRT [Computer Security Incident Response Team]

Access control

Tools

Training

Identification

The second phase deals with detecting and determining whether an incident has occurred.

Information such as error messages and log files must be gathered from various sources, including [intrusion detection systems](#) and firewalls, to make this decision.

If an incident has occurred, it should be reported as quickly as possible to give the CSIRT enough time to collect evidence and prepare for the next steps.

CSIRT members also need to be notified and begin the incident response plan process.

Containment

Once a threat has been identified, the organization must limit and prevent any further damage. There are several necessary steps to help them mitigate an incident and prevent the destruction of evidence.

1. **Short-term containment:** This aims to limit the damage as quickly as possible. It can be as simple as isolating infected machines to taking down production servers and routing all traffic to **failover** servers.
2. **System backup:** Forensic software must capture an image of affected systems as they were during the incident to preserve evidence and understand how they were compromised.
3. **Long-term containment:** This step sees the affected systems temporarily fixed to ensure they can continue to be used while rebuilding clean systems. The primary focus is for accounts or backdoors left by attackers to be removed and security patches to be installed.

Eradication or Elimination

This phase sees the removal and restoration of systems affected by the security incident.

As in all phases of the plan, documentation is crucial to determining the cost of man-hours, resources, and overall impact of the attack.

The eradication phase is also crucial to helping businesses improve their **defenses** and **fix vulnerabilities** based on the lessons they learned to make sure their systems do not get compromised again.

Recovery

This phase helps organizations carefully bring affected systems back into the production environment and ensures another incident does not occur.

Systems must be tested, monitored, and validated as they move back into production so they are not reinfected by malware or compromised.

Important decisions here include:

1. The time and date that operations are restored. System operators and owners must make the final decision based on the CSIRT's advice
2. How to test and verify that compromised systems are clean and fully functional
3. The duration that abnormal behaviors are monitored
4. Tools used to test, monitor, and validate system behavior

Lessons Learned

After an event, a lessons learned meeting should take place as soon as possible. Your report should cover:

1. When the problem was first detected, how, and by whom
2. The root cause of the incident
3. How the problem was contained and eradicated
4. Actions performed throughout the recovery process
5. Areas where the CSIRT was effective and areas for improvement
6. Suggestions and discussion around how to improve the CSIRT